

Nomination process for proposed updates to the IUCN Global Ecosystem Typology (GET)

Background

The International Union for Conservation of Nature (IUCN) Global Ecosystem Typology (GET) is a global classification system that provides a consistent framework for describing and classifying ecosystems. It distinguishes functionally different groups of ecosystems at its upper levels and compositionally distinct expressions of functional types at lower levels. Its lowest level provides a means of integrating nationally- and locally-based classifications into the global framework.

Since development in 2020, the GET has rapidly gained status as a global policy instrument. It provides the reference classification for assets in ecosystem accounts based on the UN System for Environmental Economic Accounts – Ecosystem Accounts (SEEA-EA); and is used for reporting on targets and goals of the UN Convention on Biological Diversity (e.g. for Headline Indicators of the UN Kunming-Montreal Global Biodiversity Framework).

This wide usage of the GET in connection with international policy agreements requires temporal stability in the structure and interpretation of GET classification units. On the other hand, there is a need to maintain the GET as a ‘living resource’ with updates that are responsive to advances in knowledge.

The GET Scientific Committee, comprising international experts in terrestrial, freshwater and marine ecosystems evaluates the need for specific updates to the GET, seeking external advice as needed. In evaluating proposed updates, the Committee will have regard to the trade-off between temporal consistency and structural stability of the GET and ensuring that the typology reflects current scientific knowledge. It will consider alternative types of adjustments to the GET to accommodate identified advances in knowledge while minimising the impact on users by limiting structural changes. Significant adjustments to the GET will only be endorsed if their benefits are clearly justified and outweigh the ‘costs’ in terms of temporal instability in the typology. The purpose of this proforma is to enable open input from the user community on updates to the GET and to gather the necessary information for scientifically rigorous evaluation of proposed adjustments, their benefits to the wider user community and their costs to structural stability.

Adjustment types

This process is for proposed adjustments to the Global Ecosystem Typology Levels 1 – 3, particularly Ecosystem Functional Groups (EFGs), which comprise Level 3. The process will be adapted to address adjustments to Levels 4 – 5 as their structure develops. Updates at Levels 6 are managed by local authorities and are outside the scope of this process.

Different types of adjustment to EFGs are likely to have different implications for temporal consistency of application of the GET. Four types of adjustment with increasing levels of expected impact on the temporal stability of the GET include:

- I) updates to indicative distribution maps;
- II) changes to names of GET units;
- III) changes to descriptions of GET units; and
- IV) structural changes to the classification units.

The type of adjustment should be identified and justified at the proposal stage and will be reviewed by the GET-SC in formulating its recommendation (Figure 1). The Global Ecosystem Atlas (<https://globalecosystemsatlas.org>), actively under development in a partnership led by the Group on Earth Observations (GEO) with IUCN and other parties, will provide an important source of map updates for the GET (Type I adjustments). Proposed Type IV adjustments require the strongest justification and careful consideration by the GET-SC, in conjunction with evidence of consultation and advice from specialists on the ecosystems for which adjustments are proposed (Figure 1). Adjustments with higher impact on structural stability will precipitate adjustments to multiple components of the Typology. For example, a structural adjustment creating a new Ecosystem Functional Group (EFG) will require new descriptive profiles and indicative distribution maps, as well as adjustments to those of other EFGs related to the new EFG. In evaluating the benefits and costs of proposed adjustments, the GET-SC will also consider the likely burden of updates to the information systems of user organisations.

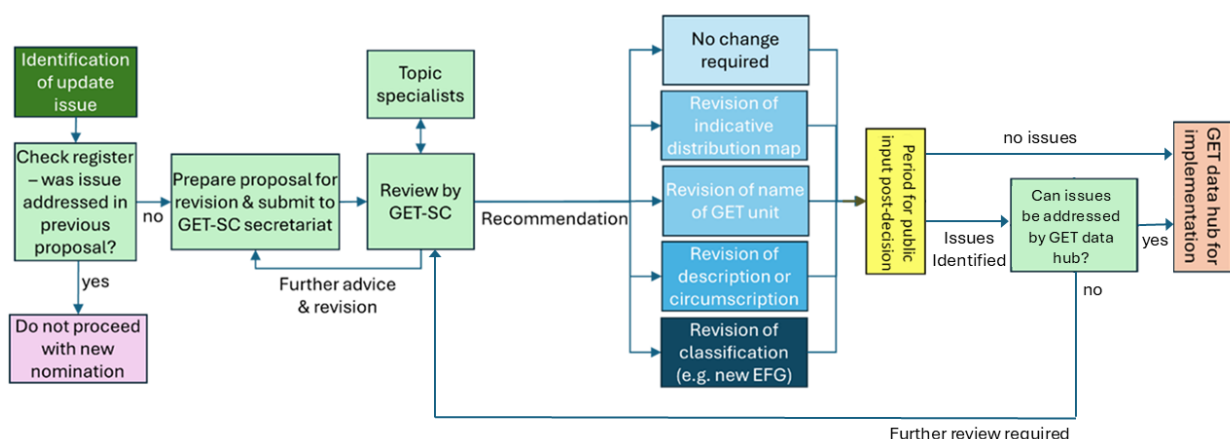


Figure 1: Overview of process for proposing and review of adjustments to the IUCN Global Ecosystem Typology.

Process for proposed adjustments

Issues relevant to updating the GET have historically been flagged from several sources: via feedback from the GET website's Feedback form, at GET workshops, and via correspondence with GET authors and IUCN staff.

A public interface for proposed adjustments to the GET is available via the current GET website (<https://global-ecosystems.org/>), with submissions collated by the GET-SC secretariat. The website includes a public register of proposed adjustments and outcomes to limit multiple proposals for the same adjustment. This should be checked by users prior to preparing and lodging proposals.

Each submission needs to describe in detail the problem with the current GET content and how it is best to be resolved, with justification for the proposed adjustment, noting strong justification is required.

Process for consideration of adjustments

The process for consideration of proposed adjustments by the Global Ecosystem Typology Scientific Committee (GET-SC) will follow the outline provided in Figure 1.

Submissions for proposed updates will be considered by the GET-SC, in two rounds each year. The GET website has details of deadline dates for submissions in each round. Proposals with insufficient justification will not be considered further and nominators will be notified.

The Committee's review may be informed by consultations with subject matter experts where needed and may involve seeking additional information from the nominator.

Table 1: Process for reviewing and implementing adjustments to the GET by adjustment type

Type of adjustment proposed	Process for review
I) updates to indicative distribution maps (typically proposed as alternative source maps);	Seek input from Global Ecosystem Atlas (GEA) content team on compliance of proposed maps with GEA data standards Seek input from original authors of EFG profile and/or other experts as required GET-SC review of merits and limitations of proposed map relative to existing map data based on: - Does map faithfully represent the relevant EFG concept - Outcome and efficacy of accuracy assessment - Extent of global coverage - Spatial resolution - Data freshness - License, copyright documentation - Availability of current and historic projections GET-SC recommendation on map with reasons

Type of adjustment proposed	Process for review
II) changes to names of GET units;	Seek input from original authors of EFG and/or other experts as required GET-SC review of need, merits (in terms of improved clarity) and limitations (in terms of user experience) of proposed name change GET-SC recommendation on updated name/s
III) changes to descriptions of GET units	Seek input from original authors of EFG and/or other experts as required GET-SC determines whether proposed adjustments alter circumscription of unit or only adds detail GET-SC reviews benefits of additional detail relative to the need for concise description GET-SC reviews how altered circumscription might benefit interpretation of the targeted unit and related units, relative to need for structural stability of the GET and implications for cross-referencing to other classifications GET-SC may revise proposed adjustments to description with input from nominator as required GET-SC recommendation on revised description
IV) structural changes to classification units (addition or deletion of classes, or significant alterations to circumscription).	GET-SC reviews justification for proposed adjustments to GET structure. Does proposed adjustment: - address a major global gap or significant improvement in representation of units consistent with the ecosystem concept, and the level of the GET - reflect a significant advance in understanding functionally distinctive ecosystem types - involve benefits (above) that outweigh likely trade-offs in terms of temporal consistency, complexity and existing cross-referencing of the GET with regard to user experience? Text, diagrams and maps drafted for the profile of any new class and necessary adjustments to profiles of existing classes, with input from nominators and/or other experts as required Review GET-SC and other topic experts Recommendation on updated structure and text/maps by GET-SC

Period for public review of proposed revisions

Following review of proposed adjustments and drafting of any required changes, the proposed adjustments will be made available for public review via the GET website (Figure 1). The period for comments or submissions on proposed adjustments will be 6 months but may be adjusted in the future. Major issues raised during the submission period will be reviewed by the GET data hub, with proposals requiring major changes or consultation referred back to the GET Scientific Committee for further consideration. Proposals that have no issues identified, or minor changes required, will be handled and implemented by the GET Data Hub.

Adjustment implementation and feedback

Accepted adjustments will be implemented by drafting, revising and reviewing maps, text, schematics and EFG structure as required.

Once adjustments have been recommended by the GET-SC, they will send materials to the GET data hub for implementation as updates to the GET website and geospatial archive.

Adjustments will be implemented in successive updates to the GET website and documented in a log for inclusion in future published versions of the GET, and their translations to other languages, ensuring that explicit version control is applied.

Recommendation outcomes will also be sent to the person or organisation who proposed the adjustment.

Appendix 1 – Proposal for an adjustment to the Global Ecosystem Typology (GET)

Contact details for proposed adjustment	Name: Xxxx Xxxxxx Organisation: XXXXXXXXXXXX Contact email: xxxxx@xxxxxx.xxx
Origin of issue: <i>[Please note if this proposed adjustment arose from a specific workshop, conference, paper, discussion etc., otherwise leave blank.]</i>	
Unit proposed for adjustment (within GET Levels 1 - 3): <i>[GET code and text label]</i>	
Other unit(s) affected by proposed adjustment: <i>[GET code and text label]</i>	
Adjustment Type <i>[Choose all that apply]</i>	☐ I. updates to indicative distribution maps; ☐ II. changes to names of GET units; ☐ III. changes to descriptions of GET units; ☐ IV. structural changes to classification units.
Summary of proposed adjustment: <i>[Identify issue with the unit and proposed resolution in summary?</i> <i>e.g. 1. Current name of EFG XXX is ambiguous and may be confused with EFG YYY. It is proposed to Rename XXX to ZZZ to improve clarity</i> <i>e.g. 2. Forests characterised by ... in regions X and Y are not readily assignable to any current EFG</i>	

<i>(explain why). It is proposed to add a new level 3 Ecosystem Functional Group to accommodate these functionally distinctive ecosystem types.]</i>	
Justification: <i>[Describe, with references, why this adjustment is essential to ensure the GET adequately represents global diversity of ecosystems at the relevant level of classification. Identify key issues affecting temporal consistency of GET applications if the adjustment is adopted. Explain why the benefits of adopting the adjustment outweigh disadvantages related to temporal inconsistency.]</i>	
Consultation: <i>[Add details of consultation undertaken in preparing this proposal, or any consultation previously undertaken, e.g. names, organisations and dates of individuals; workshops - dates, participants and outcomes etc.]</i>	
References: <i>[Add references used in text above and any other relevant sources]</i>	
Other notes:	

[Any other issues for consideration in assessing this proposal? Leave blank if not]

Appendix 2 – Examples of proposed adjustments

The following examples of proposed adjustments illustrate the range of issues that may be put forward for consideration by GET-SC for different types of adjustment (Types I-IV). With GET-SC initial commentary below each proforma, they give an indication of the justification required for each type of adjustment.

Example 1 – Level I Adjustment:

Contact details for proposed adjustment	Name: Xxxx Xxxxxx Organisation: XXXXXXXXXXXX Contact email: xxxxx@xxxxxx.xxx
Origin of issue: <i>[Please note if this proposed adjustment arose from a specific workshop, conference, paper, discussion etc., otherwise leave blank.]</i>	New project and paper documenting the world's oases (Fensham et al. 2023)
Unit proposed for adjustment (within GET Levels 1 - 3): <i>[GET code and text label]</i>	F2.8 Artesian springs and oases
Other unit(s) affected by proposed adjustment: <i>[GET code and text label]</i>	none
Adjustment Type <i>[Choose all that apply]</i>	☒ I. updates to indicative distribution maps; ☐ II. changes to names of GET units; ☐ III. changes to descriptions of GET units; ☐ IV. structural changes to classification units.
Summary of proposed adjustment: <i>[Identify issue with the unit and proposed resolution in summary?</i> <i>e.g. 1. Current name of EFG XXX is ambiguous and may be confused with EFG YYY. It is</i>	Identification of issue: Current indicative map for F2.8 is very general and low accuracy due to limited data at the time of release. Proposed adjustment: Substitute the current map with Fensham's et al. current map. The peer reviewed publication also includes information that may be helpful updates to the profile (e.g. Adding new reference and perhaps some of its content)..

<i>proposed to Rename XXX to ZZZ to improve clarity</i> <i>e.g. 2. Forests characterised by ... in regions X and Y are not readily assignable to any current EFG (explain why). It is proposed to add a new level 3 Ecosystem Functional Group to accommodate these functionally distinctive ecosystem types.]</i>	
Justification: <i>[Describe, with references, why this adjustment is essential to ensure the GET adequately represents global diversity of ecosystems at the relevant level of classification. Identify key issues affecting temporal consistency of GET applications if the adjustment is adopted. Explain why the benefits of adopting the adjustment outweigh disadvantages related to temporal inconsistency.]</i>	Since the development of the Global Ecosystem Typology, further work identifying and mapping the worlds oases has been undertaken by the Fellowship of the Spring initiative, with preliminary results published in Fensham et al. (2023). This new data will significantly improve applications of the GET for ecosystem management and reporting, outweighing any inconsistency with outdated spatial data.
Consultation: <i>[Add details of consultation undertaken in preparing this proposal, or any consultation previously undertaken, e.g. names, organisations and dates of individuals; workshops - dates, participants and outcomes etc.]</i>	Consultation has been undertaken via the Fellowship of the Spring, a multi-disciplinary team of experts and concerned stakeholders from all continents and every dryland.
References: <i>[Add references used in text above and any other relevant sources]</i>	Fensham, Roderick J., Adinehvand, Rouhollah, Babidge, Sally, Cantonati, Marco, Currell, Matthew, Daniele, Linda, Elci, Alper, Galassi, Diana M. P., de la Hera Portillo, África, Hamad, Salah, Han, Dongmei, Jawadi, Hussain Ali, Jotheri, Jaafar, Laffineur, Boris, Mohammad, Alsharifa

	Hind, Naqinezhad, Alireza, Navidtalab, Amin, Nicoll, Kathleen, Odeh, Taleb, Re, Viviana, Sanjuan, Bernard, Souza, Valeria, Stevens, Lawrence E., Tekere, Memory, Tshibalo, Ernest, Silcock, Jennifer, Webb, John, van Wyk, Braam, Zamanpoore, Mehrdad, and Villholth, Karen G. (2023). <i>Fellowship of the Spring: an initiative to document and protect the world's oases. Science of The Total Environment</i> 887 163936 163936. https://doi.org/10.1016/j.scitotenv.2023.163936
Other notes: [Any other issues for consideration in assessing this proposal? Leave blank if not]	

GET-SC commentary: This peer-reviewed study offers a major advance in the global spatial inventory of F2.8. The proposed adjustment is well justified and outweighs any inconsistency with the outdated spatial data.

Example 2 – Level II Adjustment:

Contact details for proposed adjustment	Name: Xxxx Xxxxxx Organisation: XXXXXXXXXXXX Contact email: xxxxx@xxxxxx.xxx
Origin of issue: <i>[Please note if this proposed adjustment arose from a specific workshop, conference, paper, discussion etc., otherwise leave blank.]</i>	Review of agricultural and plantation forestry ecosystems in the GET, engaging the Agricultural ecosystems ad hoc working group (Driver and Botts 2025).
Unit proposed for adjustment (within GET Levels 1 - 3): <i>[GET code and text label]</i>	T7.3 Plantations
Other unit(s) affected by proposed adjustment: <i>[GET code and text label]</i>	
Adjustment Type <i>[Choose all that apply]</i>	☐ I. updates to indicative distribution maps; ☒ II. changes to names of GET units; ☐ III. changes to descriptions of GET units; ☐ IV. structural changes to classification units.
Summary of proposed adjustment: <i>[Identify issue with the unit and proposed resolution in summary?</i> <i>e.g. 1. Current name of EFG XXX is ambiguous and may be confused with EFG YYY. It is proposed to Rename XXX to ZZZ to improve clarity</i> <i>e.g. 2. Forests characterised by ... in regions X and Y are not readily assignable to any current EFG (explain why). It is proposed to add a new level 3 Ecosystem Functional Group to accommodate these</i>	Description of issue: Name of ecosystem functional group only reflects part of the range of ecosystem types mentioned within the current descriptive profile for the group. Proposed adjustment: Name change for T7.3, from 'Plantations' to 'Plantations and woody croplands'

<i>functionally distinctive ecosystem types.]</i>	
Justification: <i>[Describe, with references, why this adjustment is essential to ensure the GET adequately represents global diversity of ecosystems at the relevant level of classification. Identify key issues affecting temporal consistency of GET applications if the adjustment is adopted. Explain why the benefits of adopting the adjustment outweigh disadvantages related to temporal inconsistency.]</i>	Plantations are areas where long-rotation perennial woody crops or trees are established and maintained for production of a variety of food and materials. The cultivated species could be trees or other woody species, either native or exotic to the locality of the plantings. Plantations include agricultural plantations, with crops such as coffee, tea and palm oil, as well as forestry plantations grown for timber and other wood products. Woody crops also include orchards (fruit or nuts) and vineyards. The name of this ecosystem functional group in Version 2.1 of the Global Ecosystem Typology is Plantations. However, based on the description it is clearly intended to include woody croplands as well as plantations. We thus propose the potential alternative name Plantations and woody croplands. This relatively minor change should improve interpretation of the GET globally, with little disruption to current usage of the GET.
Consultation: <i>[Add details of consultation undertaken in preparing this proposal, or any consultation previously undertaken, e.g. names, organisations and dates of individuals; workshops - dates, participants and outcomes etc.]</i>	Consultation has been undertaken via the Agricultural ecosystems ad hoc working group with results published in Driver and Botts (2025).
References: <i>[Add references used in text above and any other relevant sources]</i>	Driver, A. and Botts E.A. 2025. Guidelines for classifying agricultural and plantation forestry ecosystems in the IUCN Global Ecosystem Typology. Gland, Switzerland: IUCN Dupraz, C., Lawson, G., Lamersdorf, N., Papanastasis, V., Rosati, A. & Ruiz-Mirazo, J. 2018. Temperate agroforestry: the European way. doi.org/10.1079/9781780644851.0098 Chapter in: Gordon, A.M., Newman, S.M. & Coleman B.R.W. Temperate agroforestry systems. 2nd edition.

Other notes: <i>[Any other issues for consideration in assessing this proposal? Leave blank if not]</i>	

GET-SC commentary: Well justified proposal that will result in a significant improvement in clarity for users around the world, with negligible disruption to past and ongoing use of the GET.

Example 3 – Level III Adjustment:

Contact details for proposed adjustment	Name: Xxxx Xxxxxx Organisation: XXXXXXXXXXXX Contact email: xxxxx@xxxxxx.xxx
Origin of issue: <i>[Please note if this proposed adjustment arose from a specific workshop, conference, paper, discussion etc., otherwise leave blank.]</i>	Review of agricultural and plantation forestry ecosystems in the GET, engaging the Agricultural ecosystems ad hoc working group (Driver and Botts 2025).
Unit proposed for adjustment (within GET Levels 1 - 3): <i>[GET code and text label]</i>	T7.1 Annual croplands
Other unit(s) affected by proposed adjustment: <i>[GET code and text label]</i>	
Adjustment Type <i>[Choose all that apply]</i>	☒ I. updates to indicative distribution maps; ☒ II. changes to names of GET units; ☒ III. changes to descriptions of GET units; ☐ IV. structural changes to classification units.
Summary of proposed adjustment: <i>[Identify issue with the unit and proposed resolution in summary?</i> <i>e.g. 1. Current name of EFG XXX is ambiguous and may be confused with EFG YYY. It is proposed to Rename XXX to ZZZ to improve clarity</i> <i>e.g. 2. Forests characterised by ... in regions X and Y are not</i>	Identification of issue: Current description and name does not reflect the full range of cropping systems within this group. Some crops are strictly harvested destructively on an annual basis (e.g. grains), others are harvested on a longer rotation (e.g. biennial), or may include several non-destructive harvests from non-woody perennial plants within a relatively short multi-year rotation. Proposed adjustment: Broaden description and name change for T7.1 from 'Annual croplands' to 'Non-woody croplands', to reflect a broader scope of the group to include both annual and perennial herbaceous crops. This may also include updates to indicative maps for this functional group.

<i>readily assignable to any current EFG (explain why). It is proposed to add a new level 3 Ecosystem Functional Group to accommodate these functionally distinctive ecosystem types.]</i>	
Justification: <i>[Describe, with references, why this adjustment is essential to ensure the GET adequately represents global diversity of ecosystems at the relevant level of classification. Identify key issues affecting temporal consistency of GET applications if the adjustment is adopted. Explain why the benefits of adopting the adjustment outweigh disadvantages related to temporal inconsistency.]</i>	Annual croplands are areas used for growing herbaceous crops that are planted or sown and harvested within a single growing season, typically on an annual or similar short cycle (e.g. 18 months). Perennial herbaceous croplands are not explicitly addressed in v2.1 of the Global Ecosystem Typology, but from an ecosystem functioning perspective they share many characteristics with annual croplands and fit best in ecosystem functional group T7.1 (cf. alternative ecosystem functional groups). For perennial herbaceous crops, the lifespan of the plant is longer than one year, but the harvesting cycle is typically annual or a similar short cycle, or in some cases the crop is harvested continuously. The line between annual and perennial herbaceous crops is not a hard one. Some perennial species, such as cotton, casava and sugarcane, are destructively harvested and resown or replanted on an annual basis, and therefore the ecosystems in which they are produced function as annual croplands. In some areas, annual cropping activities are managed in temporal intervals of one to several years, for example, to accommodate fallow years or dry periods when there is insufficient water to support the crop (e.g. dryland cotton). The name of this ecosystem functional group in Version 2.1 of the Global Ecosystem Typology is Annual croplands. However, as noted above, from an ecosystem functioning perspective perennial herbaceous croplands share characteristics with annual croplands and fit best in this ecosystem functional group. We thus note the potential alternative name Non-woody croplands, reflecting an expansion in scope to include both annual and perennial herbaceous crops. Cropping systems for annual and perennial herbaceous crops vary, including monocropping, crop rotation (different crops grown in succession), intercropping (two or more crops grown together) and others. The target species (the species that are cultivated) are typically non-woody, with some exceptions such as cotton, which includes some woody tissue supporting its shrubby growth form. Herbaceous crops can be grown to

provide food for people, fodder for animals, or other products such as fibre or fuel.

From an ecological perspective, these are simplified systems relative to most natural ecosystems. In many cases, there is a single target plant species, and all other species of competitors or consumers are actively excluded. However, depending on the agricultural approach, some non-target species may be included or encouraged. Agroforestry practices in this ecosystem functional group include intentional integration into the cropland ecosystem.

Cropland ecosystems are often highly productive due to genetic manipulation (through breeding and in some cases genetic modification), supplementary nutrients and water, and control of weeds and pests. However, herbaceous croplands also include low productivity systems with low yields due to factors such as poor soil fertility, limited water availability and absence of supplementary nutrients. Generally, yield increases with increasing intensity of management, from low-input farms to highly intensively managed farms. Herbaceous croplands include a spectrum from small-scale subsistence production through to large commercial operations.

Annual and perennial herbaceous cropland ecosystems may be vast mono-cropped areas or they may be more varied, including mixed crops and mosaic elements such as hedgerows, patches or rows of trees, wetlands (natural or artificial), farm tracks and buildings. Where crop production areas interspersed with natural elements, they may be more ecologically complex than extensive mono-cropped areas. The presence or absence of these mosaic elements could be used to distinguish between different ecosystem types within this ecosystem functional group if they are considered to be determining factors of the composition, structure and functioning of the ecosystem. For example, the presence of hedgerows or patches of natural shrubland in a cropland might be seen as a factor that differentiates mosaic cropland ecosystem types from others in which extensive mono-cropped areas are uninterrupted by natural or semi-natural features. Mosaic agricultural ecosystem types often incorporate not only herbaceous crops but also, for example, fruit trees, pasture and some livestock. Depending on the ratio of different cultivated elements, a mosaic agricultural ecosystem might fit better in T7.2 Sown pastures or T7.3 Plantations [and woody croplands].

The features that distinguish this agricultural ecosystem functional group from others in the T7 biome are the typically herbaceous (non-woody) target species and the absence of

cultivated species that have the primary purpose of supporting grazing by herbivores (a key feature of T7.2).

Examples of ecosystems that do not match every characteristic of this ecosystem functional group but nevertheless match the description more than that of other ecosystem functional groups are:

- Cottonfields. Cotton is a woody rather than non-woody plant. With some exceptions in the tropics, it is harvested every year, with the plant removed, the field ploughed and replanted, so it is functionally closer to annual croplands than T7.3

Plantations and woody croplands.

- Pineapple fields. Pineapples are usually managed on longer than annual rotations and typically take a full year to establish before production commences. Nevertheless, the plant is herbaceous. Pineapples are harvested destructively or die after harvest (typically < 2yrs).

- Sisal plantations. Sisal is a succulent plant in the agave family, with thick fleshy leaves that contain strong fibres. Rotations are longer than other herbaceous perennials.

Examples of intermediate cases related to this ecosystem functional group, where an ecosystem shares characteristics more or less equally between this group and another, are:

- Banana plantations – intermediate between T7.1 and T7.3. Banana plants are herbaceous (very large non-woody herbs with perennial rhizomes) and from that perspective fit with T7.1. However, banana plantations have functional properties that are more like woody croplands in T7.3, such as the establishment period and rotation time (they are ploughed and replanted on longer cycles than most crops in T7.1, although shorter rotations than most woody crops in T7.3). For many users of a national ecosystem classification, banana plantations may be a more intuitive fit with T7.3. We recommend assigning banana plantations to T7.3, while recognising that they share significant characteristics with T7.1.

- Shifting cultivation in tropical forests – intermediate between T7.1 and natural forest ecosystems in T1 Tropical-subtropical forests biome. In these ecosystems, the natural forest is cleared temporarily for cropping, and then reverts to a recovering forest. If the cycles are relatively long, these ecosystems could be considered modified natural forests.

Examples of T7.1 ecosystems	Region	Type
Soybean croplands in Argentina	South America	Annual cropland

	Canola croplands in Canada	North America	Annual cropland
	Sorghum croplands in Burkina Faso	Africa	Annual cropland
	Wheat croplands in Australia	Oceania	Annual cropland; Agroforestry example
	Sunflower fields in Russia	Asia	Annual cropland
	Upland (dry) rice fields in China	Asia	Annual cropland
	Strawberry fields in the USA	North America	Perennial cropland
	Sugarcane plantations in Brazil	South America	Perennial cropland
	Cottonfields in Uzbekistan	Asia	Annual cropland; Woody plant, but best fit is T7.1
	Pineapple fields in Dominican Republic	North America	Perennial cropland
	Sisal plantations in Kenya	Africa	Perennial cropland
	Market gardens in Spain	Europe	Mixed annual and perennial
	Subsistence farming in the Ethiopian highlands	Africa	Mixed annual and perennial
Consultation: <i>[Add details of consultation undertaken in preparing this proposal, or any consultation previously undertaken, e.g. names, organisations and dates of individuals; workshops - dates, participants and outcomes etc.]</i>	Consultation has been undertaken via the Agricultural ecosystems ad hoc working group with results published in Driver and Botts (2025).		
References: <i>[Add references used in text above and any other relevant sources]</i>	Driver, A. and Botts E.A. 2025. Guidelines for classifying agricultural and plantation forestry ecosystems in the IUCN Global Ecosystem Typology. Gland, Switzerland: IUCN IUCN. 2024a. Agriculture and conservation: Living nature in a globalised world. IUCN Flagship Report Series No. 2. Gland, Switzerland: IUCN. https://portals.iucn.org/library/sites/library/files/documents/NGW-002-En.pdf Maes, J., Teller, A., Erhard, M., Grizzetti, B., Barredo, J.I., Paracchini, M.L., Condé, S., Somma, F., Orgiazzi, A., Jones, A.,		

	Zulian, A., Vallecilo, S., Petersen, J.E., Marquardt, D., Kovacevic, V., Abdul Malak, D., Marin, A.I., Czúcz, B., Mauri, A., Loffler, P., Bastrup-Birk, A., Biala, K., Christiansen, T. & Werner, B. 2018. <i>Mapping and Assessment of Ecosystems and their Services: An analytical framework for ecosystem condition</i>. Technical Report 2018 – 001. Publications office of the European Union, Luxembourg. Maes, J., Teller, A., Babí Almenar, J., Barredo, J.I., Trombetti, M., Abdul Malak, D., Paracchini M.L., Carré, A., Addamo A.M., Czúcz, B., Zulian, G., Marando F., Erhard, M., Liqueste, C., Romao, C., Polce, C., Pardo Valle, A., Jones, A., Zurbaran-Nucci, M., Nocita, M., Vysna, V., Cardoso, A.C., Gervasini, E., Magliozzi, C., Baritz, R., Barbero, M., Andre, V., Kokkoris, I.P., Dimopoulos, P., Kovacevic, V. & Gumbert, A. 2022. <i>EU-wide methodology to map and assess ecosystem condition: Towards a common approach consistent with a global statistical standard</i>. Publications Office of the European Union, Luxembourg, doi:10.2760/13048, JRC130782
Other notes: [Any other issues for consideration in assessing this proposal? Leave blank if not]	

GET-SC Commentary: This is a well justified proposal, which will improve the operability of the GET globally, and reduce ambiguity in distinguishing T7.1 from T7.3. These are likely to outweigh the relatively modest cost of adjusting existing interpretations – some users may already be applying a similar interpretation to T7.1. The proposed name may warrant further consideration of alternative options, e.g. ‘Short-rotation croplands’ encompassing the woody crop exceptions such as cotton.

Example 4 – Level IV Adjustment:

Contact details for proposed adjustment	Name: Xxxx Xxxxxx Organisation: XXXXXXXXXXXX Contact email: <u>XXXXX@XXXXXX.XXX</u>
Origin of issue: <i>[Please note if this proposed adjustment arose from a specific workshop, conference, paper, discussion etc., otherwise leave blank.]</i>	
Unit proposed for adjustment (within GET Levels 1 - 3): <i>[GET code and text label]</i>	T1.2 – Tropical/Subtropical dry forests and thickets
Other unit(s) affected by proposed adjustment: <i>[GET code and text label]</i>	
Adjustment Type <i>[Choose all that apply]</i>	☒ I. updates to indicative distribution maps; ☒ II. changes to names of GET units; ☒ III. changes to descriptions of GET units; ☒ IV. structural changes to classification units.
Summary of proposed adjustment: <i>[Identify issue with the unit and proposed resolution in summary?</i> <i>e.g. 1. Current name of EFG XXX is ambiguous and may be confused with EFG YYY. It is proposed to Rename XXX to ZZZ to improve clarity</i> <i>e.g. 2. Forests characterised by ... in regions X and Y are not readily assignable to any current EFG (explain why). It is proposed to add a new level 3 Ecosystem Functional Group to accommodate these</i>	Identification of issue: The current functional group encompasses a large range of variation along both a water balance gradient and a soil texture/fertility gradient. Ecosystem properties such as leaf phenology and vegetation structure (notably height) vary along these gradients. The relationships are best understood in the Cerrado regions of South America. Proposed adjustment: Split T1.2 into two Functional Groups or subgroups at Level 3 of the GET: semideciduous forests vs. deciduous forests and thickets.

<i>functionally distinctive ecosystem types.]</i>	
Justification: <i>[Describe, with references, why this adjustment is essential to ensure the GET adequately represents global diversity of ecosystems at the relevant level of classification. Identify key issues affecting temporal consistency of GET applications if the adjustment is adopted. Explain why the benefits of adopting the adjustment outweigh disadvantages related to temporal inconsistency.]</i>	The greater amount of soil water found in deciduous forests than in semideciduous forests was associated with clay soils and greater availability of organic matter (Kilca et al., 2009). In Cerrado regions, deciduous forests are commonly associated with more fertile soils (Bueno et al., 2018), which have lower acidity and higher levels of bases and organic matter (Silva and Scariot 2004; Bueno et al., 2018; de Souza et al., 2024). On the other hand, semideciduous forests are found in soils with lower natural fertility (V% = 11.3 to 53.22) and are more acidic (pH = 4.14 to 5.38) (Souza et al., 2012), presenting similarities in terms of soil properties (e.g., chemical and textural) to humid forest soils, which are characterized by a reduced CEC, acidic pH and low V%.
Consultation: <i>[Add details of consultation undertaken in preparing this proposal, or any consultation previously undertaken, e.g. names, organisations and dates of individuals; workshops - dates, participants and outcomes etc.]</i>	Consultation has been undertaken within IBGE (Brazil) and a workshop is planned to address this issue. Further consultation required across other range countries
References: <i>[Add references used in text above and any other relevant sources]</i>	Bueno ML, Dexter KG, Pennington RT, et al. (2018) The environmental triangle of the Cerrado Domain: Ecological factors driving shifts in tree species composition between forests and savannas. <i>J Ecol.</i> 106: 2109–2120. https://doi.org/10.1111/1365-2745.12969 de Souza, A.P., Corrêa, G.R., Arruda, D.M. <i>et al.</i> Edaphic gradient shapes the structure and composition of vegetation in the forest-cerrado ecotone in north of minas gerais, Brazil. <i>Braz. J. Bot</i> 47, 1147–1161 (2024). https://doi.org/10.1007/s40415-024-01036-5 Kilca, R. V., Schiavini, I., Araújo, G. M., & Fagg, J. M. F. (2009). Edaphic and structural differences between two

	seasonal forests in the Cerrado biome. Neotropical Biology and Conservation, 4, 150–163. Silva, L.A. and Scariot, A. (2004) Composição e estrutura da comunidade arbórea de uma floresta estacional decidual sobre afloramento calcário no Brasil central. Árvore (Viçosa, Brasil), 28, 69-75. https://doi.org/10.1590/S0100-67622004000100009
Other notes: <i>[Any other issues for consideration in assessing this proposal? Leave blank if not]</i>	

GET-SC Commentary: This proposal makes a case for splitting an existing EFG into two based on variation in properties related primarily to an edaphic gradient. However, further work is needed before GET-SC can make a recommendation on the proposal. First the evidence comes only from one region in a globally distributed EFG – do the relationships hold in other regions (Asia, Africa, Oceania) or is one of the proposed types unique to South America? Evidence of further consultation in other regions is required. Second, more information is needed on the ecosystem properties that distinguish the two groups – the information on edaphic properties and leaf phenology is foundational, but what other functional properties (of vegetation, fauna, physical environment) might distinguish the proposed groups as a basis for full descriptive profiles? Third, what mapping data are available or needed to characterise the different distributions of the proposed groups? Finally, what are the major global benefits of this proposed adjustment that outweigh user costs of adapting their information and reporting systems for a globally distributed EFG? Can the issue be addressed via a less disruptive adjustment, e.g. revising the descriptive profile of T1.2 (Type III adjustment) to address the variation in ecosystem properties along the gradients, with formal recognition as different types at lower levels of the GET?